

Department of Computer Science
Guru Nanak Dev University, Amritsar
MCA (TYP) CBEGS– Sem II
CSL 4099: THEORY OF COMPUTATION

Time Allowed: 1 hour

Max Marks: 30

Note: All questions are compulsory. Each question carries equal marks.

- Q1. Let G be the grammar $S \rightarrow 0B|1A$, $A \rightarrow 0|0S|1AA$, $B \rightarrow 1|1S|0BB$. For the string 00110101. Find the leftmost derivation, right most derivation and the derivation tree. (7)
- Q2. Convert the following **CFG** into **CNF**:
 $S \rightarrow AB|a$, $A \rightarrow aA|\epsilon$, $B \rightarrow bB|\epsilon$. (8)
- Q3. Explain the closure properties of Regular language classes. Construct a regular expression for the language containing all strings over $\{0,1\}$ that have at least one '0' and end with '1'. (4,3)
- Q4. Construct a DFA to accept a string containing 0's as multiple of 3. (8)
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